

Understanding the Evidence Lower Bound (ELBO) in Variational Autoencoders (VAEs)

1 Objective of VAEs

The main goal of a Variational Autoencoder (VAE) is to learn a generative model that can approximate the true data distribution $p(x)$. Directly maximizing the likelihood of the data $p(x)$ is often intractable, especially when dealing with complex distributions and latent variables.

2 Bayes' Theorem

To understand the ELBO, it's useful to look at Bayes' theorem:

$$p(x) = \int p(x|z)p(z)dz$$

Here:

- $p(x|z)$ is the likelihood of the data given the latent variables z .
- $p(z)$ is the prior distribution over the latent variables.

The integral computes the marginal likelihood of the data $p(x)$, which is usually difficult to compute due to the intractable integration over all possible latent variables.

3 Variational Inference

To make the problem tractable, we use variational inference. Instead of directly calculating $p(x)$, we approximate it by introducing a variational distribution $q(z|x)$ to estimate the posterior $p(z|x)$.

4 The ELBO

The ELBO is derived from the relationship between the true likelihood $p(x)$ and the variational distribution $q(z|x)$. It can be expressed as:

$$\text{ELBO} = \mathbb{E}_{q(z|x)}[\log p(x|z)] - \text{KL}(q(z|x)||p(z))$$

Breaking this down:

- **First Term** $\mathbb{E}_{q(z|x)}[\log p(x|z)]$: This is the expected log-likelihood of the data given the latent variables. It represents how well the model can reconstruct the input data from the latent variables and serves as the reconstruction loss.
- **Second Term** $-\text{KL}(q(z|x)||p(z))$: This is the Kullback-Leibler (KL) divergence between the variational distribution $q(z|x)$ and the prior distribution $p(z)$. It acts as a regularization term that encourages the learned distribution to be close to the prior distribution.

5 Maximizing the ELBO

By maximizing the ELBO, we effectively:

- Improve the reconstruction of the input data (first term).
- Ensure that the learned latent representation conforms to the prior (second term), leading to a well-structured latent space.

6 Relationship to Marginal Likelihood

Maximizing the ELBO also provides a lower bound on the marginal likelihood of the data:

$$\log p(x) \geq \text{ELBO}$$

Thus, maximizing the ELBO indirectly helps improve the model's ability to generate data similar to the training set, as it works towards maximizing $p(x)$.

7 Summary

In summary, the ELBO is a crucial component in training VAEs. It balances the quality of data reconstruction with the regularization of the latent space, facilitating efficient learning and enabling the generation of new data samples. By maximizing the ELBO, VAEs effectively optimize their parameters to achieve a good approximation of the true data distribution while maintaining a well-structured latent space.